Physics-guided characterisation-time reduction for second-life LFP cells: how few cycles does PyBaMM's reaction-limited SEI need?

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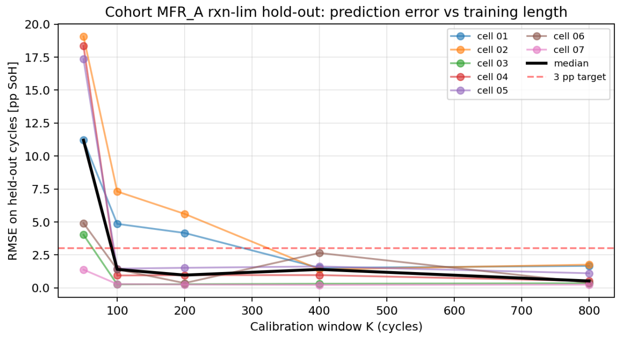
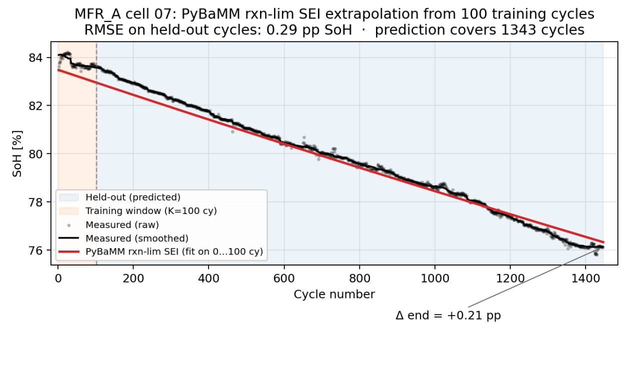
Motivation. Second-life repurposing of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost: pack builders must estimate remaining useful life (RUL) with no beginning-of-life data and limited laboratory cycling time. We ask: *how many measured cycles does a physics-based degradation model need before it can predict the remaining life of a used cell?*

Method. We calibrate PyBaMM's reaction-limited SEI degradation model (`Prada2013` chemistry, isothermal 25 °C) against the measured fade slope of a used cell using only its first K cycles, then run the calibrated model to the full measured horizon and score against the held-out cycles K–N. A single scalar (SEI exchange current density j_{SEI}) is fit per cell by bisection on $\log_{10}(j_{SEI})$; per-cell open-circuit voltage, initial state-of-health, and ohmic resistance are constrained from HPPC and OCV–SoC characterisation. We sweep $K \in \{50, 100, 200, 400, 800\}$ across seven MFR_A-cohort cells with clean long-trajectory records ($N > 1000$ cycles each) that pass an upstream reaction-limited-shape filter.

Result. Prediction RMSE on held-out cycles drops from a median of 11.2 pp SoH at $K = 50$ to 1.4 pp at $K = 100$ and 0.5 pp at $K = 800$ (Fig. 1a). At $K = 400$, all 7/7 cells achieve $RMSE_{test} < 3$ pp SoH: **400 vs. ≈1200 measured cycles per cell (3× reduction, ≈4 months of test-lab time saved at 0.25 C).** Five of seven cells already converge at $K = 100$ with end-of-trajectory error below 0.4 pp (Fig. 1b) — a **12× reduction** for that sub-cohort. The remaining two exhibit a post-formation recovery transient (positive apparent slope across cycles 50–200) and require $K \geq 400$; this signal itself classifies cells for correct calibration windowing.



(a) Held-out RMSE vs. training window K; red dashed = 3 pp target.



(b) K=100-cycle fit predicts remaining 1345 cycles of cell 07 to 0.29 pp SoH RMSE.

Scope & PyBaMM usage. The result holds only within the reaction-limited SEI regime. Cells flagged by an upstream shape filter (non-monotonic-early, three-regime, or accelerating anomalies) are excluded — such shapes may reflect LAM domination (where PyBaMM's stress-driven LAM lacks saturation and overfits) or data-processing artefacts (batch-transition discontinuities in cycle records). PyBaMM (`v26.4.3`) is the calibration target and extrapolation engine; per-cell characterisation is mapped into the `Prada2013` parameter set via `build_pybamm_parameters`, and bisection over $\log_{10}(j_{SEI})$ uses cached 10-cycle PyBaMM runs. Scripts released at `https://github.com/VKC-Turno/PyBaMM-Neural-ODE-RUL`.

References. [1] V. Sulzer et al., *J. Open Res. Softw.* 9(1) 14–21 (2021). [2] M. Prada et al., *J. Electrochem. Soc.* 160(4) A616–A628 (2013).